Playing with Probability: An Introduction to Stochastic Processes

SCIENCE PROGRAM
COURSE OUTLINE
WINTER 2026

General Information.

Discipline:	Mathematics
Course code:	201-SNP-AB
Ponderation:	2-2-2
Credits:	2
Prerequisites:	201-SN1-RE (Probability and Statistics) 201-SN2-RE (Calculus 1)

Objective:

- 0GNF: Consolidate one's scientific culture in a scientific field.

> Your teacher will give you his/her schedule and availability.
> *Students are strongly advised to seek help promptly from their teacher if they encounter difficulties in the course.*

Introduction. *Playing with Probability* is an optional course in the Science program. It is usually taken in the third or fourth semester.

The primary purpose of the course is the attainment of Objective 0GNF ("Consolidate one's scientific culture in a scientific field."). The following quote summarizes well the broad subject of stochastic processes.

"*Stochastic, or random, processes is the dynamic side of probability. The material appeals to those who like applications and to those who like theory. It is both excellent preparation for future study, as well as a terminal course, in the sense that we do not have to tell students to wait until the next class or the next year before seeing the good stuff. This* is *the good stuff! Stochastic processes, as a branch of probability, speaks in the language of rolling dice, flipping coins, and gambling games, but in the service of applications as varied as the spread of infectious diseases, the evolution of genetic sequences, models for climate change, and the growth of the World Wide Web.*" (Robert P. Dobrow)

This course contributes to Exit Profile goals 1, 4 and 5 of the Science program.

Teaching Methods. This is a 60-hour course meeting three times a week (two classes of 90 minutes and one class of 60 minutes) for a total of four hours per week. Classes are primarily lectures, but other teaching methods may be used (for example: labs, problem-solving sessions, class discussions, and assigned reading for independent study). Regular homework requiring an average of two hours per week should be expected. Students are responsible for all homework problems assigned by the teacher.

Software. Students will build on their knowledge of R (a freely available language and environment for statistical computing and graphics) acquired in Probability and Statistics (201-SN1-RE) to explore different probability models. Those without a personal computer or laptop will have access to college computers (for example, in the Math Study Area) where the required software is installed.

Reference. There is no required textbook for this course. Good references for the course material are

Introduction to Stochastic Processes with R,
by Robert P. Dobrow (Wiley, 2016)
Introduction to Probability Models, Ninth Edition,
by Sheldon M. Ross (Academic Press, 2007)

A set of exercises will be provided by your teacher.

Course Costs.
A scientific (non-graphing, non-programmable) calculator is required for the lectures and for all the assessments. A suitable model (SHARP EL-531) is available at the bookstore for around $24.

Evaluation Plan. A student's Final Grade will be calculated by the following distribution:

-	Class Tests:	50%	(two written in-class tests)
-	Project:	25%	(involving the software R)
-	Minor Assessments:	25%	(see outline addendum)

The minor assessments may consist of labs, quizzes, assignments, or other types of evaluations. The breakdown of minor assessments is detailed in an appendix that is distributed to students along with this course outline.
Students will choose the topic of their project based on a list of potential topics provided by the teacher. A teacher may also allow students to choose a different project topic (following the teacher's guidance).
The Final Evaluation Task in this course consists of the two class tests and the project, which together cover all elements of the competency.

Course Content. The topics listed below follow the presentation and section numbering from the reference text by Dobrow. Your teacher may supplement this list with additional material from other sources during the semester. Regular work done as the course progresses should make it easier for you to master the course.

General Probability

- Review of basic probability and discrete random variables
- Continuous random variables

Markov Chains

- Introduction, definition, notation (Section 2.1)
- Transition matrix
- Examples such as gambler's ruin and random walks (Section 2.2)
- The n-step transition matrix and long-term behavior (Section 2.3-2.4)
- Simulation using R (Section 2.5)
- Limiting distributions and stationary distributions (Section 3.1-3.2)
- Recurrent and transient states (Section 3.3)
- Simple random walks in one, two, three dimensions
- Absorbing chains (Section 3.8)

Branching Processes

- Introduction (Section 4.1)
- Mean generation size (Section 4.2)
- Probability generating functions (Section 4.3)
- Extinction probability theorem, simulation in R (Section 4.4)

Poisson Process

- Definition using infinitesimal probabilities (Section 6.3)
- Definition using the Poisson distribution (Section 6.1)
- Definition using the Exponential distribution (Section 6.2)
- Equivalence of definitions (sections 6.1, 6.2, 6.3)
- Thinning and supersposition of Poisson processes (Section 6.4)
- Connection with the Uniform distribution (Section 6.5) - if time permits

More on Random Walks - if time permits

- Statement of the Law of Large Numbers, simulations
- Statement of the Central Limit Theorem, simulations
- Stating convergence of rescaled random walks to Brownian motion, simulations (Section 8.2)

Brownian Motion - if time permits

- Processes with stationary increments and independent increments
- Definition of Brownian motion (Section 8.1)
- Invariance principle (Section 8.2)
- Last Zero Standing (Corollary 8.2) and Fluctuations in Coin Tossing (Example 8.9)
- Brownian motion with drift, Brownian bridge (definitions only)

Other Resources.

Math Website.
http://departments.johnabbott.qc.ca/departments/mathematics

Math Study Area. Located in H-200A and H-200B; the common area is usually open from 8:30 to 17:30 on weekdays as a quiet study space. Computers and printers are available for math-related assignments. It is also possible to borrow course materials when the attendant is present.

Math Help Centre. Located in H-216; teachers are on duty from 8:30 until 15:30 to give math help on a drop-in basis.

Academic Success Centre. The Academic Success Centre, located in H-117, offers study skills workshops and individual tutoring.

College Policies.

Policy No. 7 - IPESA, Institutional Policy on the Evaluation of Student Achievement: http://johnabbott.qc.ca/ipesa.

Religious Holidays (Article 3.2.13 and 4.1.6). Students who wish to miss classes in order to observe religious holidays must inform their teacher of their intent in writing within the first two weeks of the semester.

Student Rights and Responsibilities: (Article 3.2.18). It is the responsibility of students to keep all assessed material returned to them and/or all digital work submitted to the teacher in the event of a grade review. (The deadline for a Grade Review is 4 weeks after the start of the next regular semester.)

Student Rights and Responsibilities: (Article 3.3.6). Students have the right to receive graded evaluations, for regular day division courses, within two weeks after the due date or exam/test date, except in extenuating circumstances. A maximum of three (3) weeks may apply in certain circumstances (ex. major essays) if approved by the department and stated on the course outline. For evaluations at the end of the semester/course, the results must be given to the student by the grade submission deadline (see current Academic Calendar). For intensive courses (i.e.: intersession, abridged courses) and AEC courses, timely feedback must be adjusted accordingly.

Academic Procedure: Academic Integrity, Cheating and Plagiarism (Article 9.1 and 9.2). Cheating and plagiarism are unacceptable at John Abbott College. They represent infractions against academic integrity. Students are expected to conduct themselves accordingly and must be responsible for all of their actions.

College definition of Cheating: Cheating means any dishonest or deceptive practice relative to examinations, tests, quizzes, lab assignments, research papers or other forms of evaluation tasks. Cheating includes, but is not restricted to, making use of or being in possession of unauthorized material or devices and/or obtaining or providing unauthorized assistance in writing examinations, papers or any other evaluation task and submitting the same work in more than one course without the teacher's permission. It is incumbent upon the department through the teacher to ensure students are forewarned about unauthorized material, devices or practices that are not permitted.

College definition of Plagiarism: Plagiarism is a form of cheating. It includes copying or paraphrasing (expressing the ideas of someone else in one's own words), of another person's work or the use of another person's work or ideas without acknowledgement of its source. Plagiarism can be from any source including books, magazines, electronic or photographic media or another student's paper or work.

OBJECTIVES	STANDARDS
Statement of the competency	**General Performance Criteria for the Competency as a Whole**
Consolidate one's scientific culture in a scientific field. (0GNF).	• Appropriate use of terminology • Appropriate use of necessary computer tools
Elements of the Competency	**General Performance Criteria**
1. Identify concepts specific to the chosen field.	• Appropriate characterization of relevant concepts • Accurate association of concepts to their field of application
2. Solve problems specific to the chosen field.	• Accurate application of the relevant concepts, laws and principles • Rigorous application of a procedure adapted to the problem • Clear presentation of problem-solving steps • Accurate interpretation of results
3. Demonstrate the contribution of the field to understanding scientific issues.	• Clear delimitation of a relevant problem involving scientific issues • Accurate application of concepts, laws and principles relevant to the problem • Establishment of relevant connections between the field and the scientific issues studied